



Immersed Boundary Method for Study of Flapping Wing Mechanism in Low Reynolds' Number Regimes

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Abstract

The featherwing beetle *Paratuposa placentis* is among the smallest known flying insects, operating at Reynolds numbers of $Re \approx 60$. Its hindwings consist of a central *rachis* fringed with long bristles rather than a continuous membrane. At these scales, viscous forces dominate and the inter-bristle flow structure governs aerodynamic force production.

This study validates the **Immersed Boundary Method (IBM)** implementation in OpenFOAM against canonical benchmarks – flow over a cylinder, over a sphere, and a flapping plate – as a precursor to simulating the bristled wing geometry of *P. placentis*. IBM avoids costly body-fitted remeshing while accurately imposing no-slip conditions on moving boundaries via an SDF-based force approach.

Problem Statement

Existing computational studies of bristled wings use either *porous-medium approximations*, which cannot resolve inter-bristle flow, or explicit 2D *cylinder-array models* limited to planar assumptions. Three-dimensional resolved simulations exist but employ wavelet-adaptive solvers and have not been applied under the documented kinematics of *P. placentis*.

No study has applied IBM within OpenFOAM to this problem with species-specific kinematics. The present work targets this gap by validating IBM and establishing it as a tool for simulating bristled wing aerodynamics – quantifying lift, drag, and inter-bristle leakage at the relevant Re .

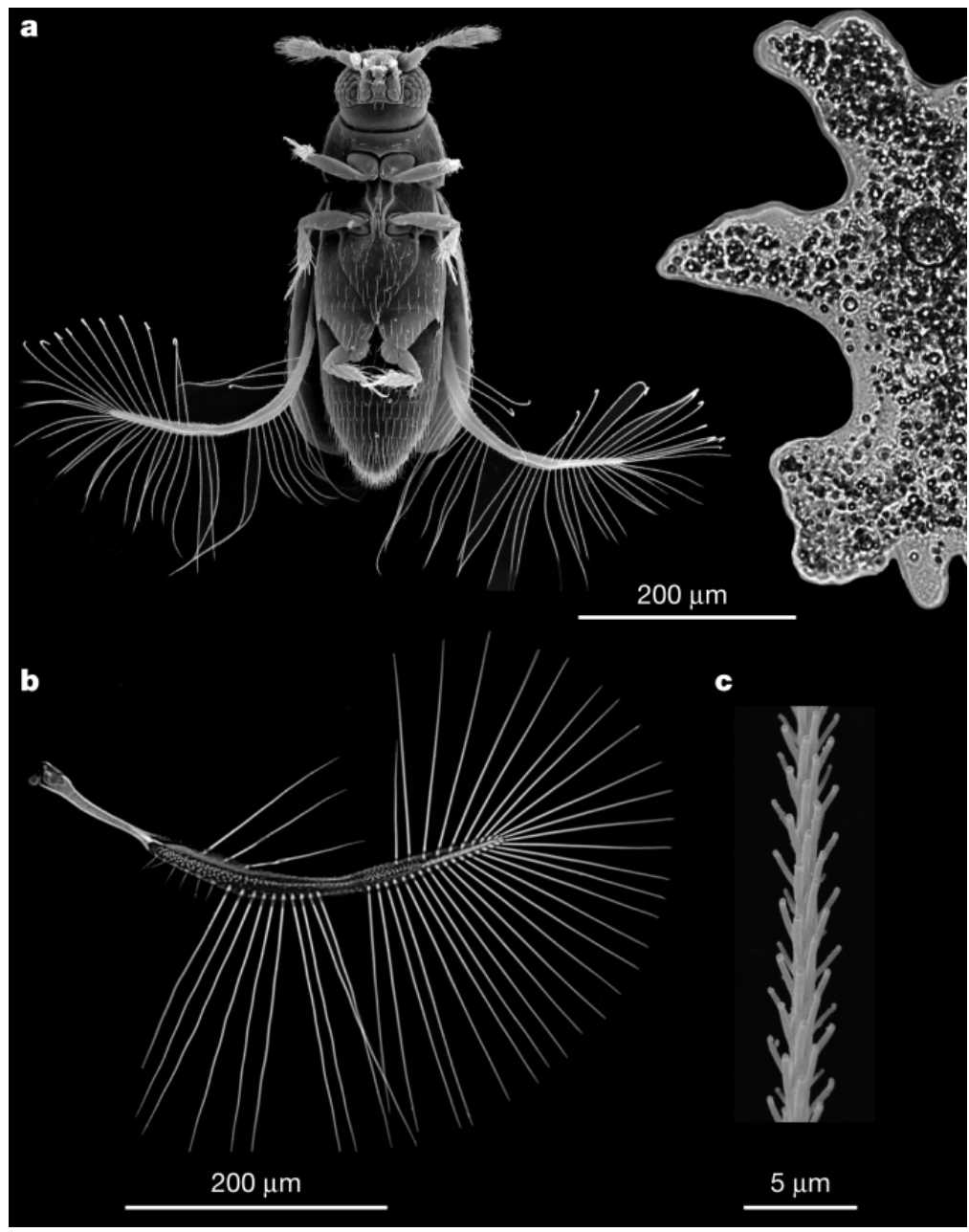


Figure 1. Scanning electron microscopy images showing (a) relative size of *P. placentis* (left) and *A. proteus* (right) (b) wing of *P. placentis* and (c) part of a seta.

The Finite Volume Method in OpenFOAM

OpenFOAM solves the incompressible Navier-Stokes equations using the **Finite Volume Method (FVM)**. The domain is discretised into a fixed Cartesian mesh of control volumes. Fluxes of momentum and mass are computed at each face using Gauss's theorem:

$$\frac{\partial u}{\partial t} + (u \cdot \nabla) \cdot u = -\nabla p + \nu \nabla^2 u + f$$
$$\nabla \cdot u = 0$$

The **PIMPLE** algorithm (merged PISO-SIMPLE) handles pressure-velocity coupling. Time integration uses a second-order backward scheme. The IBM body force f is applied as a source term directly in the momentum equation, replacing the need for boundary-conforming mesh generation around moving surfaces.

Immersed Boundary Method

The original algorithm designed by C. Peskin distributes boundary forces to the fluid using a regularised *Dirac delta* function $\delta(x)$, spreading the no-slip condition over several grid cells around the immersed surface:

$$f(x) = \int F(s) \cdot \delta(x - X(s)) \cdot ds$$

This smearing introduces force diffusion errors, and the effective boundary location depends on grid resolution. The method is sensitive to the choice of delta kernel width.

SDFIBM (Zhang *et al.*) replaces the delta function with the **Signed Distance Function (SDF)** $\phi(x)$, which stores the signed perpendicular distance from each fluid cell to the nearest solid surface. Cells with $\phi < 0$ are inside the body; the body force is applied as a direct velocity correction in a narrow band around $\phi = 0$:

$$\phi(x) = \pm \text{dist}(x, \partial s)$$
$$u^*(x) = u_{\text{body}}; \text{ where } \phi(x) \leq 0$$
$$f(x) = \frac{u_{\text{body}} - u_{\text{fluid}}}{\Delta t}$$

This gives a sharp, geometry-consistent boundary without kernel tuning. The SDF is recomputed each timestep for moving bodies using a fast marching method, making it naturally suited to large-displacement motion such as flapping wings.

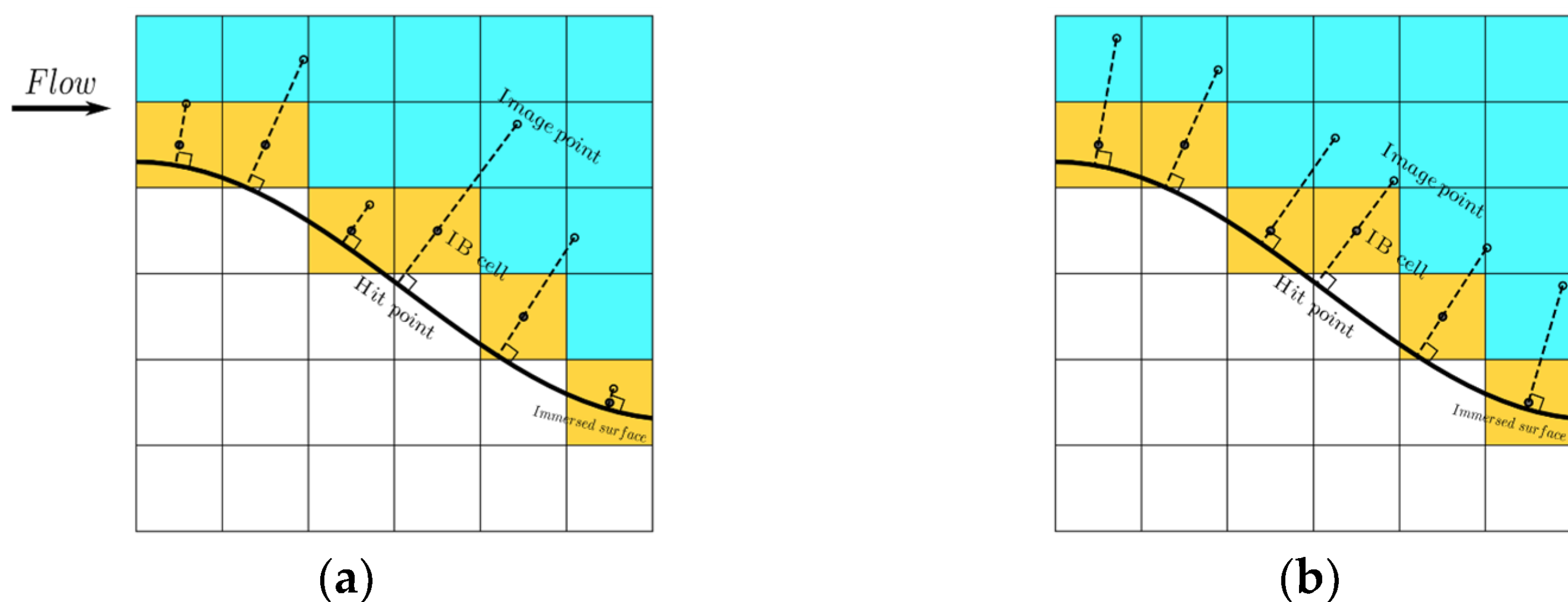


Figure 2. Distribution of body force among surrounded mesh cells

Simulation Results

Preliminary Cases: Cylinder and Sphere moving inside a fluid

Re	ν	U_∞	D
10	$10^{-4} m^2/s$	$0.01 m/s$	$0.1 m$
No. of Cells	Domain Size	Δx	Δt
156,400	$35D \times 20D$	$3.33 mm$	$0.01 s$
$C_{d,sim}$	$C_{d,th}$	Error	
3.01	3.06	1.5%	

Table 1. Case Parameters and Results for Cylinder Simulation

Re	ν	U_∞	D
10	$10^{-4} m^2/s$	$0.01 m/s$	$0.1 m$
No. of Cells	Domain Size	Δx	Δt
3,146,000	$35D \times 20D \times 20D$	$10 mm$	$0.01 s$
$C_{d,sim}$	$C_{d,th}$	Error	
4.16	4.15	0.2%	

Table 2. Case Parameters and Results for Sphere Simulation

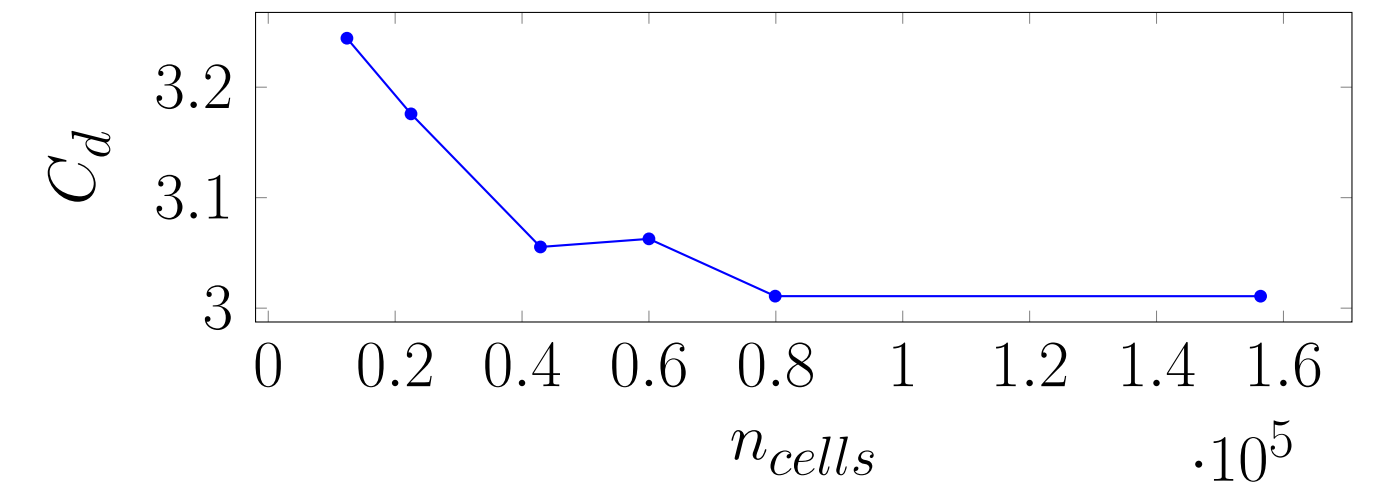


Figure 3. Mesh Convergence for Cylinder

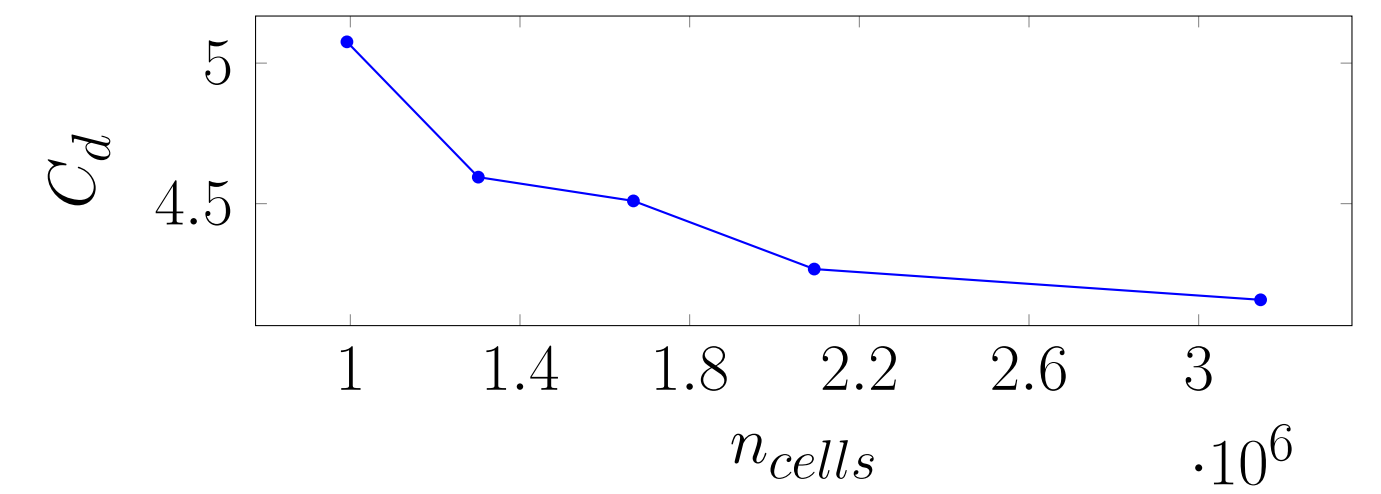


Figure 4. Mesh Convergence for Sphere

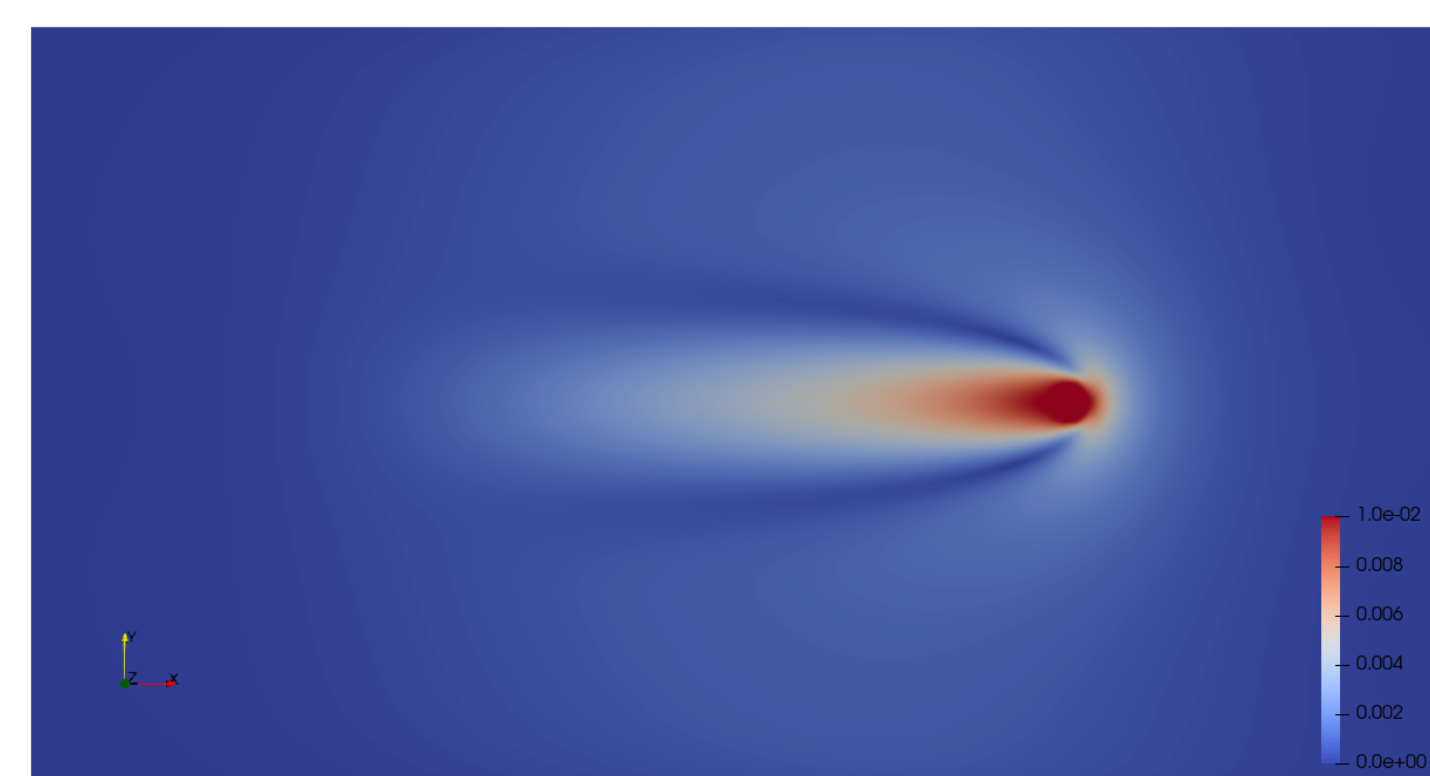


Figure 5. Velocity Contours for Cylinder

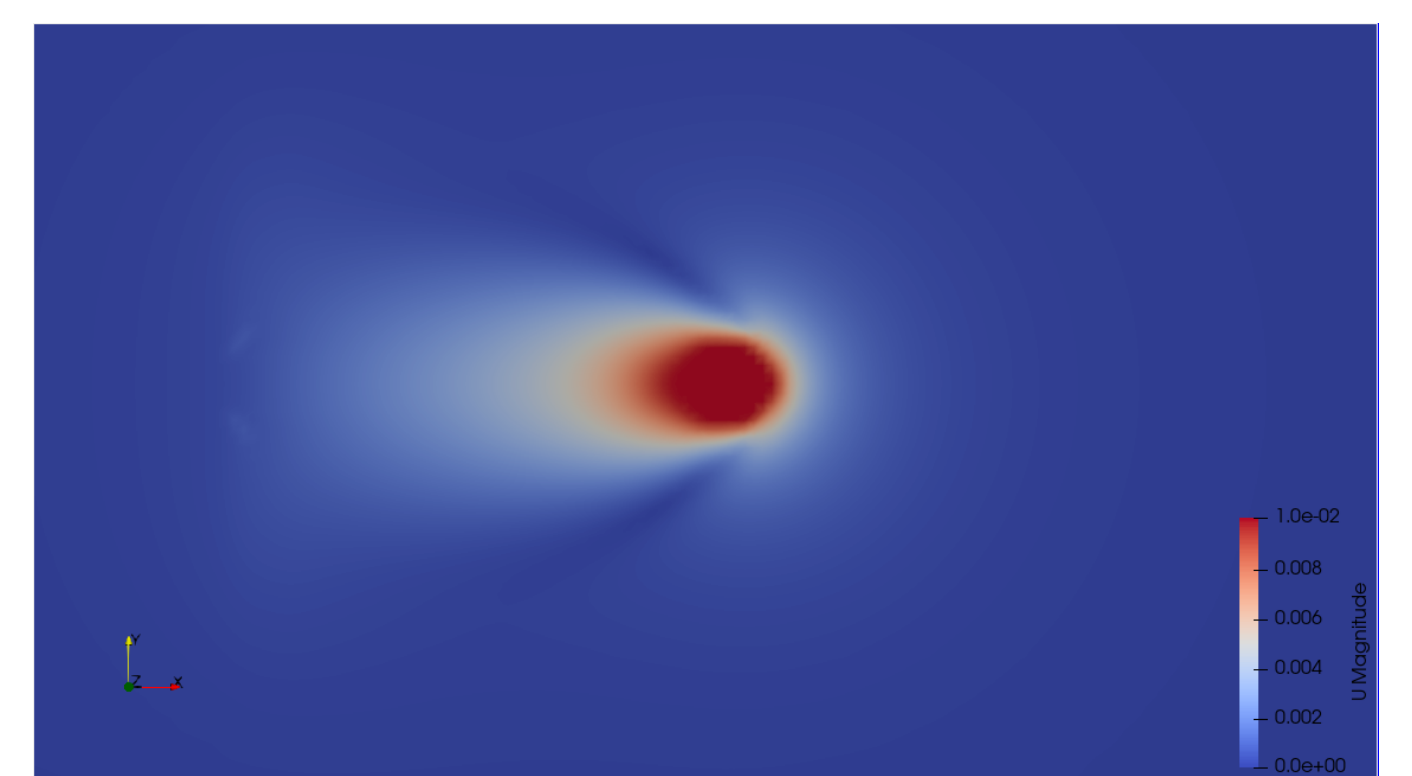


Figure 6. Velocity Contours for Sphere

Main Case: Flat plate flapping inside a fluid

The flapping motion can be mathematically defined by:

$$x(t) = -h_a - h_a \cos(2\pi \cdot \frac{t}{T})$$
$$\theta(t) = 90 \text{ deg} - \alpha_a \sin(2\pi \cdot \frac{t}{T})$$

Re	ν	U_{ref}	D
100	$10^{-4} m^2/s$	$0.1 m/s$	$0.1 m$
No. of Cells	Domain Size	Δx	Δt
500000	$6D \times 3D$	$0.6 mm$	$0.01 s$

Table 3. Case Parameters for Flapping Plate

h_a	α_a	T
$0.15 m$	45 deg	$9.42 s$

Table 4. Motion Parameters for Flapping Plate

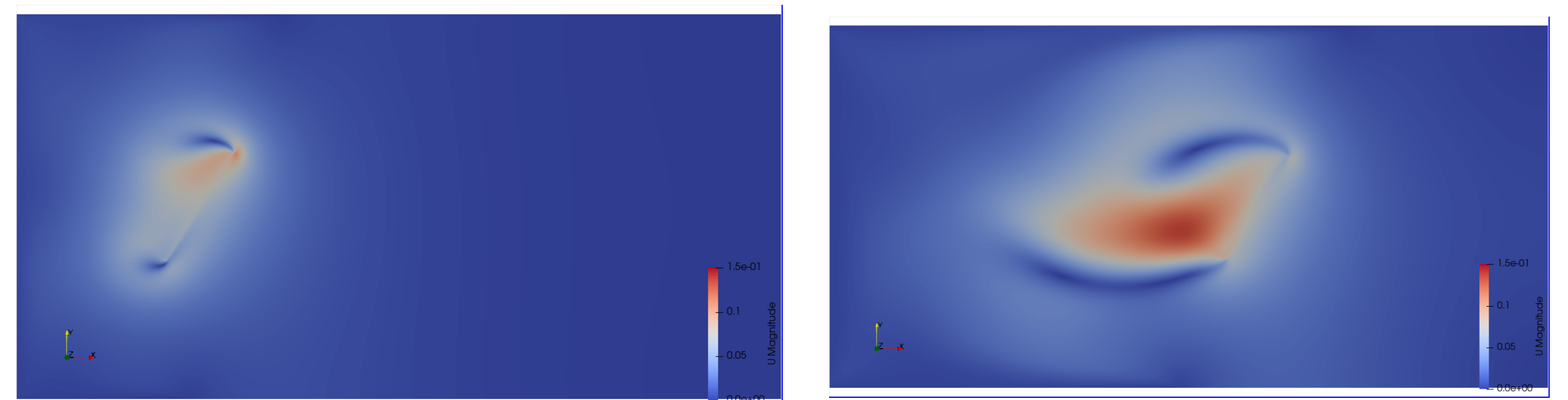


Figure 7. Velocity Contours for Flapping Plate at (a) $t/T = 0.25$ (b) $t/T = 0.75$

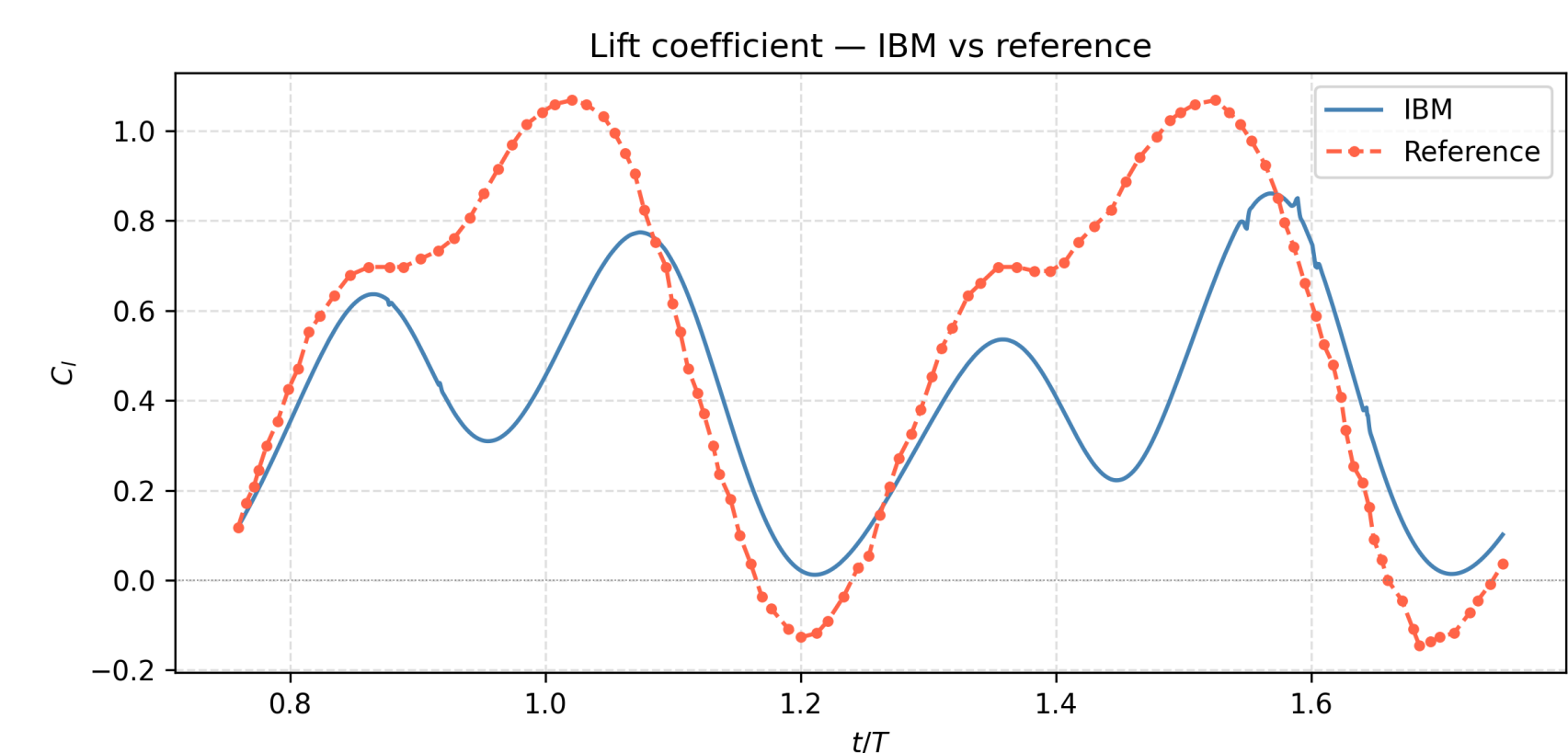


Figure 8. Variation of Coefficient of Lift v/s Time

Discussion

Here we can see that the simulation qualitatively matches the reference values – we can observe two peaks in 1 half time interval (half flap) – but the values differ in magnitude. This can be attributed to a coarse mesh and observation of early fluid flow where the flow hasn't reached a stable state yet. Other possible reasons are under inspection.

References

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- [2] Charles S. Peskin. The immersed boundary method. *Acta Numerica*, 11:479–517, 2002.
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